Tutorial 4 & 5

Installation of CloudSim Simulator
Demonstration and program implementation on CloudSim Simulator

Observation:

Tutorial 4/5:

CloudSim

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CloudSim is an open-source simulation framework used to model and analyze cloud computing environments. It enables researchers, students, and developers to study how cloud systems behave without requiring access to actual cloud infrastructure such as AWS, Azure and Google Cloud.

CloudSim is implemented in Java and is commonly executed using Java IDEs such as Eclipse IDE or IntelliJ IDEA. It provides flexible platform for simulating cloud components like data centers, hosts, virtual machines, cloudlets (tasks), and scheduling policies.

With CloudSim users can:

→ Model one or more data centers
→ Create hosts with configurable CPU, memory, storage and bandwidth.
→ Launch Virtual Machines (VMs)
→ Submit cloudlets representing computational tasks.
→ Apply scheduling and allocation policies.
→ Measure execution time, cost and resource utilization

CloudSim simulates the following -

1. Data Centers - A data center consists of multiple physical hosts that provide computing resources.

2. Hosts - Hosts are physical servers with CPU cores, RAM, storage and network bandwidth.

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3. Virtual Machines (VMs) - VMs are created hosts and are allocated a share of the host's resources.

4. Cloudlets - Cloudlets are user tasks or jobs that are executed on VMs.

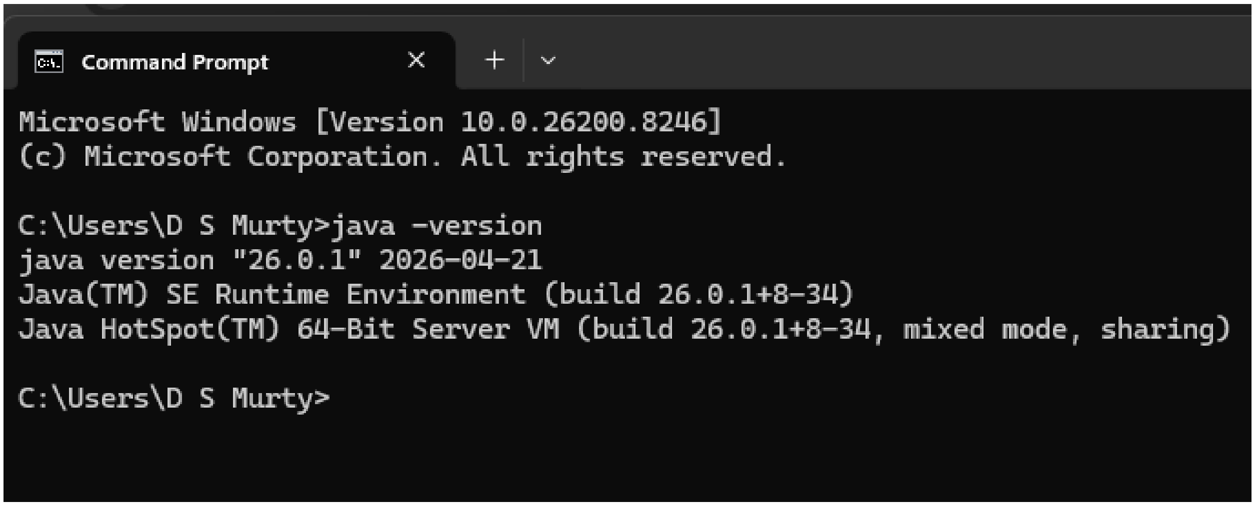
5. Brokers - A broker acts as an intermediary that submits VMs and cloudlets to the data center and manages their execution.

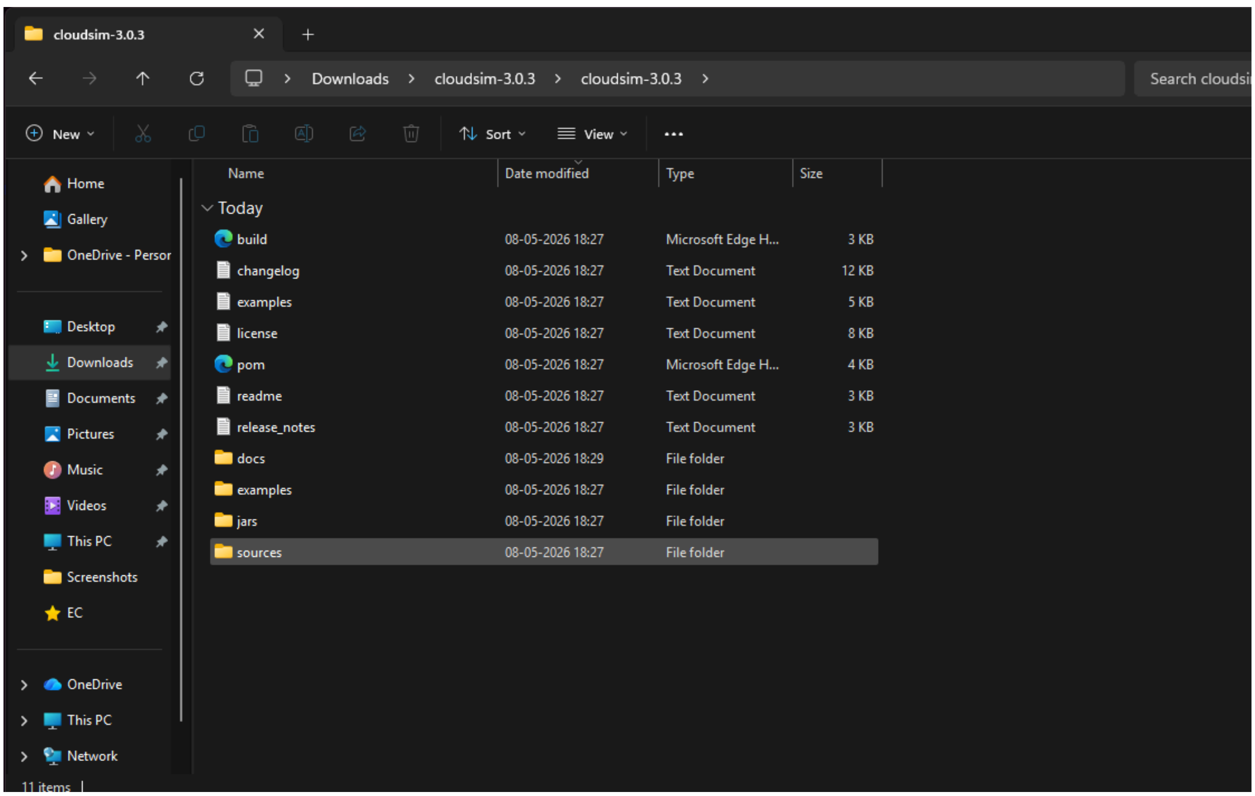
Key Features -

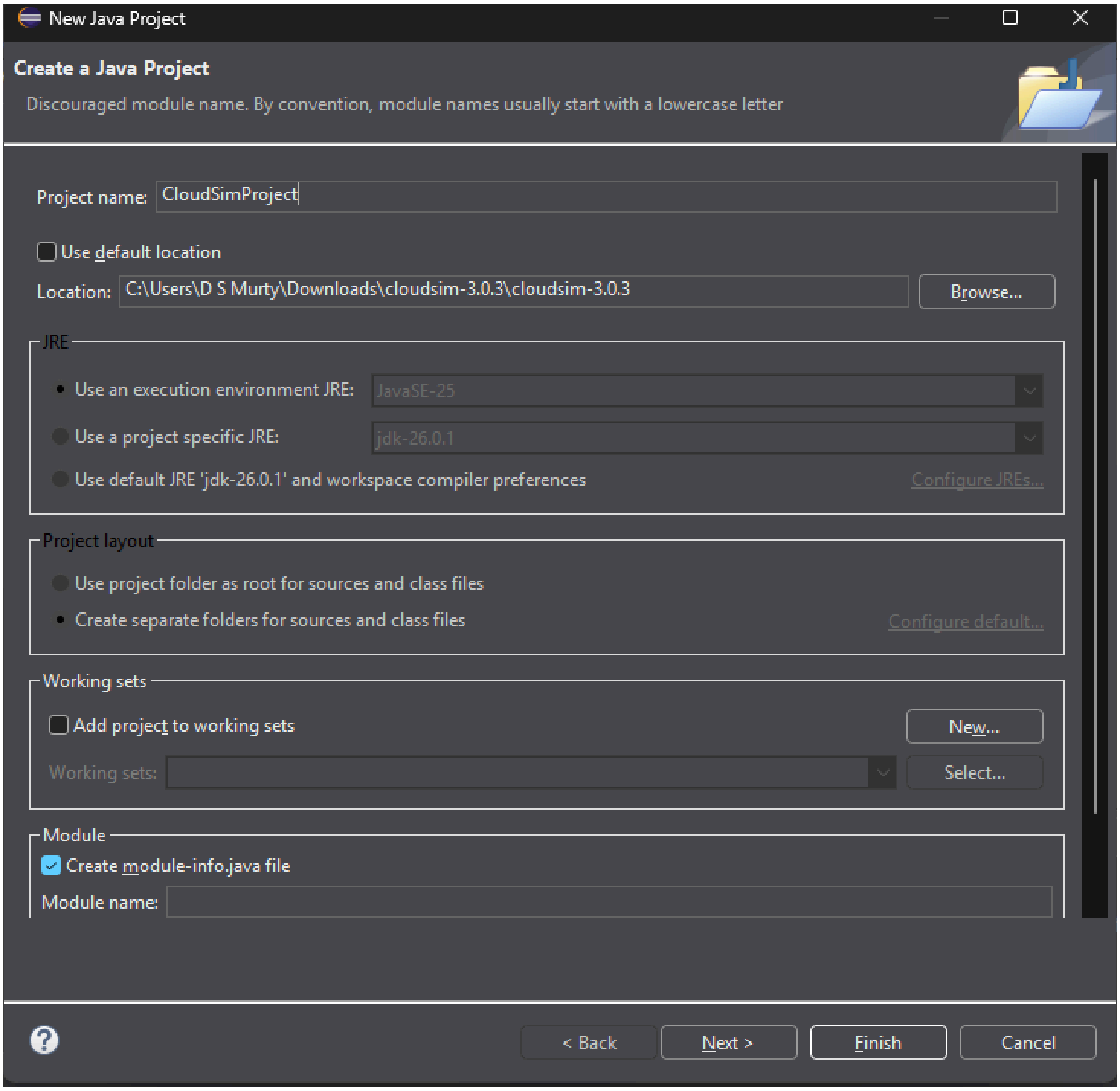
→ Supports large-scale cloud simulations.
→ Allows customization of scheduling algorithms.
→ Models energy-aware and cost-aware environments.
→ Enables performance comparison under different configurations.
→ Provides detailed execution statistics.

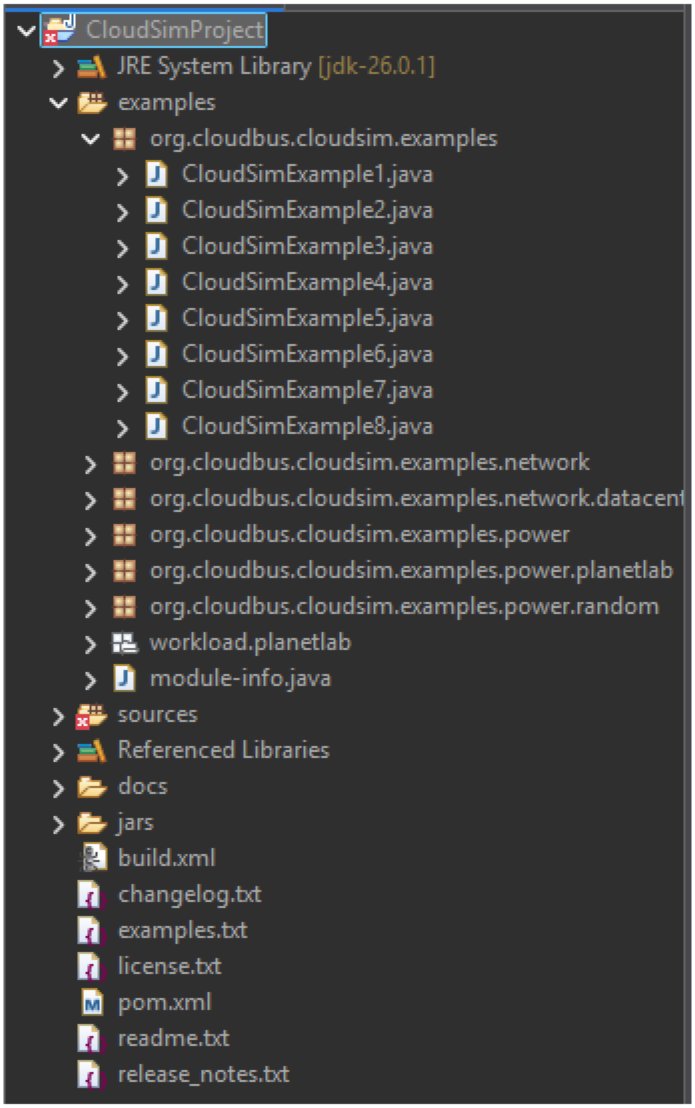
Applications -
* Evaluating VM scheduling algorithms.
* Testing load balancing strategies.
* Studying energy-efficient resource allocation.
* Comparing cloud service configurations.
* Teaching cloud computing concepts.

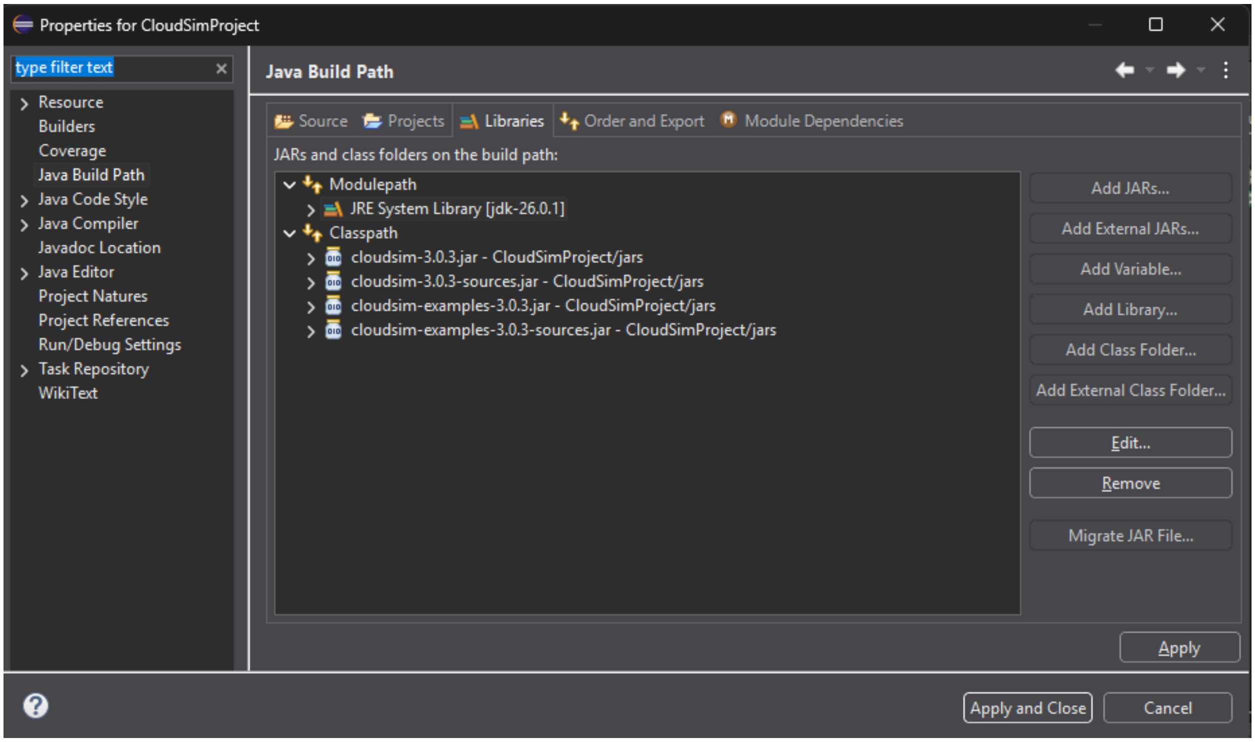
Screenshots:

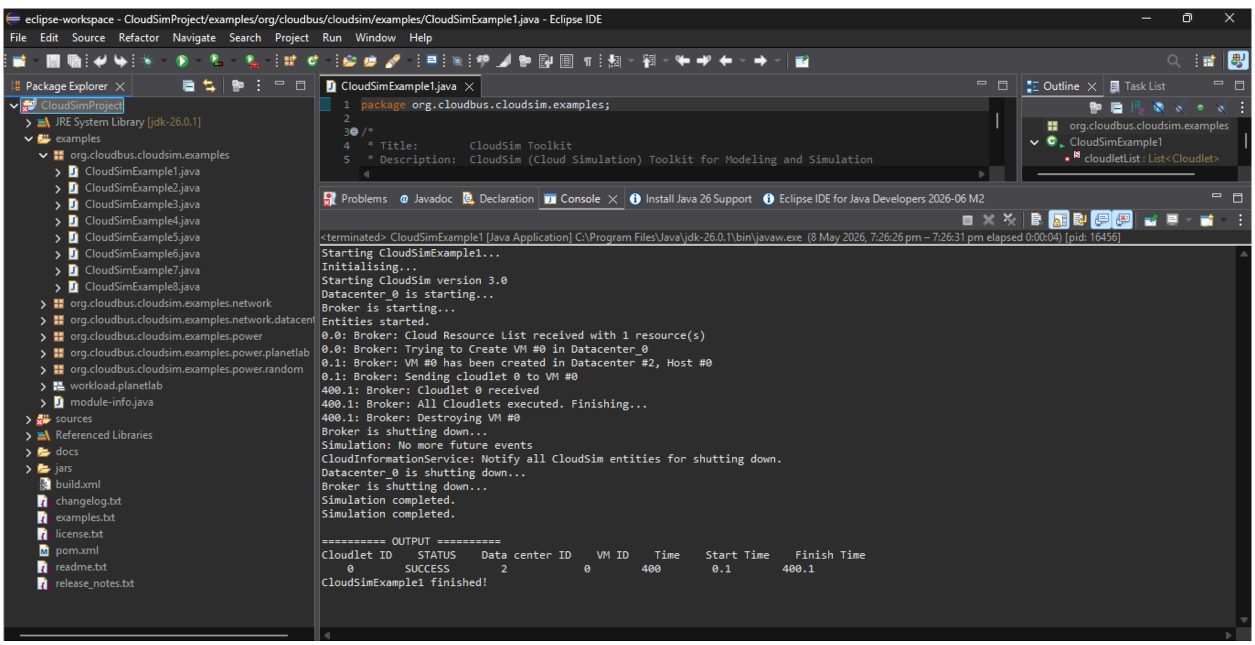





New Java Project
Create a Java Project
Discouraged module name. By convention, module names usually start with a lowercase letter
Project name: CloudSimProject
Use default location
Location: C:\Users\D S Murty\Downloads\cloudsim-3.0.3\cloudsim-3.0.3
Browse...
JRE
Use an execution environment JRE: JavaSE-25
Use a project specific JRE: jdk-26.0.1
Use default JRE 'jdk-26.0.1' and workspace compiler preferences
Configure JREs...
Project layout
Use project folder as root for sources and class files
Create separate folders for sources and class files
Configure default...
Working sets
Add project to working sets
New...
Working sets:
Select...
Module
Create module-info.java file
Module name:
< Back
Next >
Finish
Cancel


CloudSimProject
JRE System Library [jdk-26.0.1]
examples
org.cloudbus.cloudsim.examples
CloudSimExample1.java
CloudSimExample2.java
CloudSimExample3.java
CloudSimExample4.java
CloudSimExample5.java
CloudSimExample6.java
CloudSimExample7.java
CloudSimExample8.java
org.cloudbus.cloudsim.examples.network
org.cloudbus.cloudsim.examples.network.datacent
org.cloudbus.cloudsim.examples.power
org.cloudbus.cloudsim.examples.power.planetlab
org.cloudbus.cloudsim.examples.power.random
workload.planetlab
module-info.java
sources
Referenced Libraries
docs
jars
build.xml
changelog.txt
examples.txt
license.txt
pom.xml
readme.txt
release_notes.txt


Properties for CloudSimProject
type filter text
Resource
Builders
Coverage
Java Build Path
Java Code Style
Java Compiler
Javadoc Location
Java Editor
Project Natures
Project References
Run/Debug Settings
Task Repository
WikiText
Java Build Path
Source
Projects
Libraries
Order and Export
Module Dependencies
JARs and class folders on the build path:
Modulepath
JRE System Library [jdk-26.0.1]
Classpath
cloudsim-3.0.3.jar - CloudSimProject/jars
cloudsim-3.0.3-sources.jar - CloudSimProject/jars
cloudsim-examples-3.0.3.jar - CloudSimProject/jars
cloudsim-examples-3.0.3-sources.jar - CloudSimProject/jars
Add JARs...
Add External JARs...
Add Variable...
Add Library...
Add Class Folder...
Add External Class Folder...
Edit...
Remove
Migrate JAR File...
Apply
Apply and Close
Cancel


eclipse-workspace - CloudSimProject/examples/org/cloudbus/cloudsim/examples/CloudSimExample1.java - Eclipse IDE
File Edit Source Refactor Navigate Search Project Run Window Help
Package Explorer
CloudSimProject
JRE System Library [jdk-26.0.1]
examples
org.cloudbus.cloudsim.examples
CloudSimExample1.java
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CloudSimExample7.java
CloudSimExample8.java
org.cloudbus.cloudsim.examples.network
org.cloudbus.cloudsim.examples.network.datacent
org.cloudbus.cloudsim.examples.power
org.cloudbus.cloudsim.examples.power.planetlab
org.cloudbus.cloudsim.examples.power.random
workload.planetlab
module-info.java
sources
Referenced Libraries
docs
jars
build.xml
changelog.txt
examples.txt
license.txt
pom.xml
readme.txt
release_notes.txt
CloudSimExample1.java
package org.cloudbus.cloudsim.examples;
/*
* Title: CloudSim Toolkit
* Description: CloudSim (Cloud Simulation) Toolkit for Modeling and Simulation
Outline
Task List
org.cloudbus.cloudsim.examples
CloudSimExample1
cloudletList : List<Cloudlet>
Problems
Javadoc
Declaration
Console
Install Java 26 Support
Eclipse IDE for Java Developers 2026-06 M2
<terminated> CloudSimExample1 [Java Application] C:\Program Files\Java\jdk-26.0.1\bin\javaw.exe (8 May 2026, 7:26:26 pm – 7:26:31 pm elapsed 0:00:04) [pid: 16456]
Starting CloudSimExample1...
Initialising...
Starting CloudSim version 3.0
Datacenter_0 is starting...
Broker is starting...
Entities started.
0.0: Broker: Cloud Resource List received with 1 resource(s)
0.0: Broker: Trying to Create VM #0 in Datacenter_0
0.1: Broker: VM #0 has been created in Datacenter #2, Host #0
0.1: Broker: Sending cloudlet 0 to VM #0
400.1: Broker: Cloudlet 0 received
400.1: Broker: All Cloudlets executed. Finishing...
400.1: Broker: Destroying VM #0
Broker is shutting down...
Simulation: No more future events
CloudInformationService: Notify all CloudSim entities for shutting down.
Datacenter_0 is shutting down...
Broker is shutting down...
Simulation completed.
Simulation completed.
========== OUTPUT ==========
Cloudlet ID STATUS Data center ID VM ID Time Start Time Finish Time
0 SUCCESS 2 0 400 0.1 400.1
CloudSimExample1 finished!